

Joemon Jesudas

Gen AI Engineer | ML Engineer | MLOps Engineer

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SKILLS

Primary Skills : Python, SQL, Machine Learning, Generative AI, LLM Application Development, RAG (Retrieval-Augmented Generation), LangChain, Azure OpenAI, Azure AI Foundry, Azure Document Intelligence, PGVector, Vector Databases, Embeddings, Prompt Engineering, FastAPI, MCP Frameworks

Secondary Skills: Azure Cloud Services, Docker, Kubernetes, MLflow, Grafana, PySpark, Apache Airflow, Power BI, Streamlit, DVC, Git, GitHub Actions, Azure Data Lake Storage, Angular.js

Professional Experience

May 2022 - PRESENT

Allianz Services

ML/Gen AI Engineer

- Architected and deployed end-to-end RAG pipelines using LangChain, Azure OpenAI, and PGVector, enabling semantic search over 10,000+ insurance policy documents with sub-second retrieval and contextually accurate LLM-generated responses.
- Developed a RAG-based liability prediction system that analyzes accident descriptions and recommends liability decisions for claim handlers, improving decision consistency by 30% and reducing average manual review time.
- Designed and implemented document extraction pipelines using Azure Document Intelligence, performing structured data extraction, validation checkpoints, and ingestion into PGVector for downstream RAG and semantic search workflows.
- Engineered chunking strategies, embedding pipelines, and vector indexing workflows using PGVector and OpenAI embeddings to optimize retrieval relevance and reduce hallucination in RAG applications.
- Built custom prompt templates, few-shot examples, and guardrail logic (content filtering, output validation, hallucination checks) for high-accuracy LLM responses across document understanding and claims processing tasks.
- Developed and deployed traditional machine learning models (Gradient Boosting, Random Forest, XGBoost) for insurance claims handling, resulting in savings of more than 1.2 million GBP for the client.
- Built RESTful APIs using FastAPI to serve ML model predictions and LLM-generated responses, enabling seamless integration with downstream applications and front-end interfaces.
- Built web-based ML and GenAI applications using Streamlit, enabling claim handlers and business stakeholders to interact with model predictions and LLM-generated insights through intuitive interfaces.
- Designed and scheduled cron jobs for automated data extraction from Azure Data Lake, deploying them in Kubernetes clusters for reliable, production-grade data orchestration.
- Data Integration: Integrated data from various sources (social media platforms, communication logs, etc...) to perform comprehensive social network analysis.
- Implemented model monitoring using MCP and Claude Desktop integrated with Evidently AI to generate natural-language insights and proactively identify data drift or model performance anomalies.
- Developed interactive Grafana dashboards to monitor LLM and ML pipeline health, combining telemetry, token usage, latency metrics, SQL-based aggregations, and operational KPIs.
- Containerized LLM and ML solutions using Docker and orchestrated deployments on Kubernetes via Helm charts, ensuring reproducible, scalable, and environment-consistent production deployments.

- Implemented CI/CD pipelines with GitHub Actions, automating build, test, and deployment workflows for ML/LLM applications, reducing deployment cycle time significantly.
 - Managed data versioning and experiment tracking using DVC and MLflow, ensuring reproducibility across model iterations and RAG configuration changes.
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Orglens

Dec 2021 - March 2022

Social Network Analyst - Internship

- Social Network Analysis: Conducted social network analysis to identify key influencers, community structures, and information flow within networks.
 - Network Visualization: Created visual representations of social networks using NetworkX, to illustrate complex relationships and structures.
 - Data Integration: Integrated data from various sources (social media platforms, communication logs, etc...) to perform comprehensive social network analysis.
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Cognizant Technology Solutions

Dec 2018 - May 2020

Programmer Analyst

- Managed SAP security operations including role provisioning, and authorization maintenance across multiple business units, ensuring compliance with internal security policies.
 - Managed 120+ tickets per month through ServiceNow, improving issue resolution efficiency by 40% and strengthening customer satisfaction.
 - Monitored SAP system alerts, background jobs, RFC connections, and system logs, proactively identifying and resolving performance issues.
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Publications

- "Model Monitoring With Grafana and Dynatrace: A Comprehensive Framework For Ensuring ML Model Performance" - <http://dx.doi.org/10.21474/IJAR01/18684>
 - "Claims Negotiation Amount Predictor: A Gradient Boosting Regression Based Machine Learning Solution with Model Monitoring" - <http://www.ijrar.org/papers/IJRAR23D1678.pdf>
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Education & Academics

Masters in Computer Science

Rajagiri college of social sciences(Autonomous)
MG University

2020 - 2022

CGPA : 3.96/5.0 (B+)

Bachelors in Computer Application

Kristu Jyoti College of Management and Technology
MG University.

2015 - 2018

CCPA : 7.62/10.0 (B)

Awards and Certifications

- Certified in Azure AI Engineer Associate - AI102
- Certified in Azure Fundamentals (Microsoft) - AZ900
- Certification in MLOps - Cloudxlabs
- Certification in AIOps - iNeuron
- Allianz Best Performer Award, Feb - 2026
- Allianz Best All-rounder Q2 - 2025
- Allianz Best All-rounder Q1 - 2025
- Allianz Delivery Excellence Champion - 2023
- Allianz Best Performer Award, April - 2023
- Allianz Best Performer Award, July - 2023
- Allianz Best Performer Award, Dec - 2022